

Osmium MIMU48XC + BMBT4444

User Guide

Revision 1.0

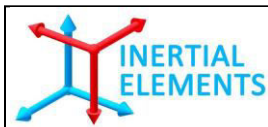
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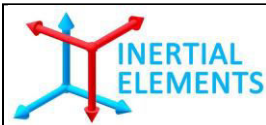
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Revision History

Revision	Revision Date	Updates
1.0	13 August 2018	Initial version



Purpose & Scope

This document describes different features about MIMU48xC and BMBT444.

Refer following documents also for specification details:

1. [Datasheet –ICM-20948](#)
2. [32-bit AVR Microcontroller Specification](#)

Refer the following document for Communication Protocol

1. [Integration Guide](#)

Refer the following document for Stepwise Data and Packet Acknowledgement:

1. [Application Note](#)

MIMU48XC

MIMU48XC = MIMU4844 + 10 pin Connector

The **Osmium MIMU48XC** is a variant of massive inertial sensory array module MIMU4844, with a 10-pins connector. The connector allows access to one set each of UART and SPI IOs of the micro-controller. The connector also allows access to power IOs. This arrangement allows integration of hardware accessories like Bluetooth, Memory card etc. in form of extension board (or shield) with the sensor array module. Access to power IOs enables integration of battery power management with MIMU48XC. The device is configured to work as a standalone ZUPT-aided inertial navigation system and as a displacement & heading change sensor. Python Script is available for communication.

MIMU48XC, when used with BMBT4444, is capable of storing data in micro SD card, operating on battery power and, most importantly, wireless communication with the application platform. Mating of MIMU48XC and BMBT4444 results in an easy-to-use integrated device, which is capable of communicating with the application platform wirelessly. These features are in addition to all the features of MIMU4844. These features and capabilities make MIMU48XC a suitable platform for carrying out research in motion sensing by using Sensor Fusion and Array Signal Processing methods. MIMU48XC, like MIMU4844, is an easy to use and highly configurable hardware platform, serves the needs for niche applications, such as gait analysis, autonomous robotics, 3D motion capture, indoor positioning, Structure from Motion (SfM) etc.

10-pins connector description:

UART: RX and TX

SPI: MOSI, MISO, SCK, GND

Power pins: VDD (5V out), GND, VIN (For enabling battery operability)

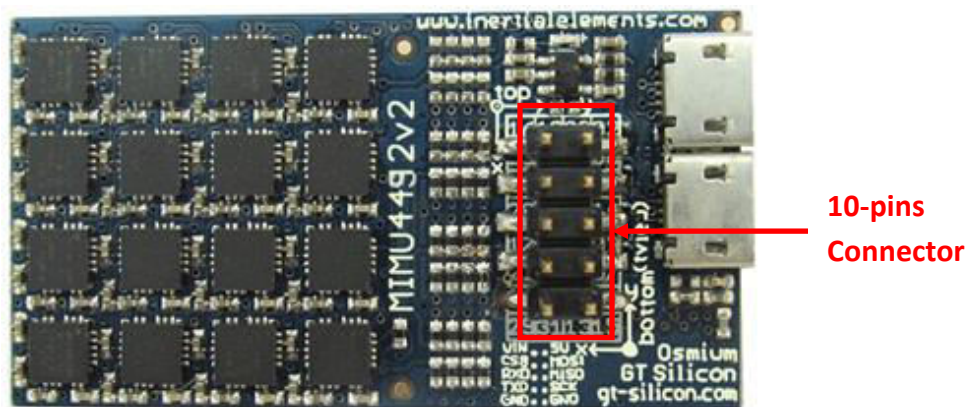


Figure 1: The Osmium MIMU48XC

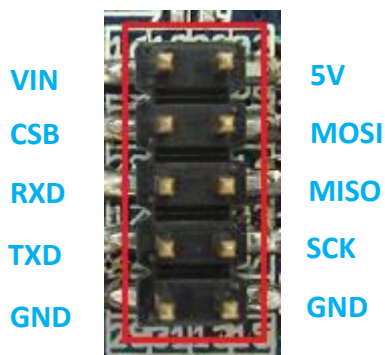


Figure 2: The 10-pins male connector on MIMU48XC

In order to use MIMU48XC without BMBT4444, one must short VIN and 5V on MIMU48XC board, using a jumper. This is like using MIMU48XC as MIMU4844.

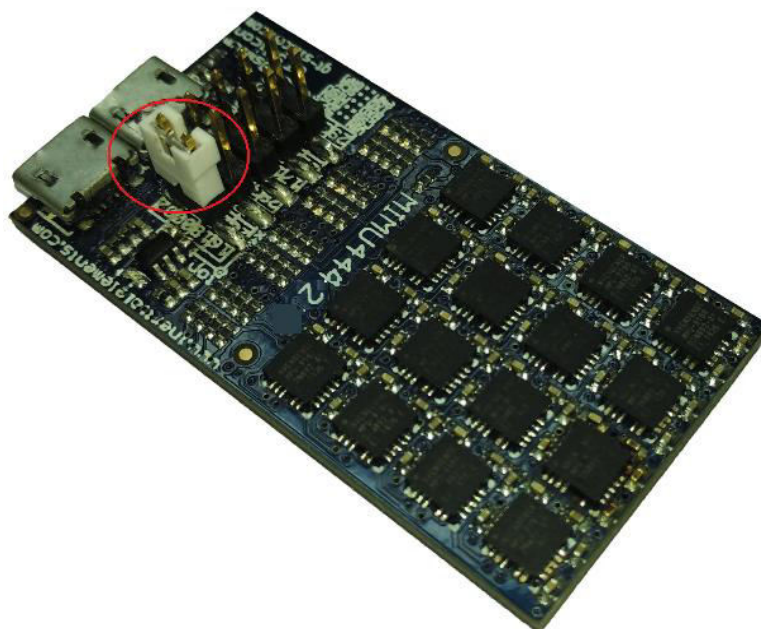


Figure 3: MIMU48XC with a jumper for shorting VIN and 5V pins

Please go through the technical specifications and integration guide of MIMU4844/MIMU48XC for details.

BMBT4444

The **BMBT4444** serves the purpose of extension board for MIMU4844. When used with MIMU4844, it adds on-off switch, wireless communication interface (Bluetooth), data storage (Micro SD Card) and battery (Li-ion rechargeable) as another powering option to the inertial sensor array module.

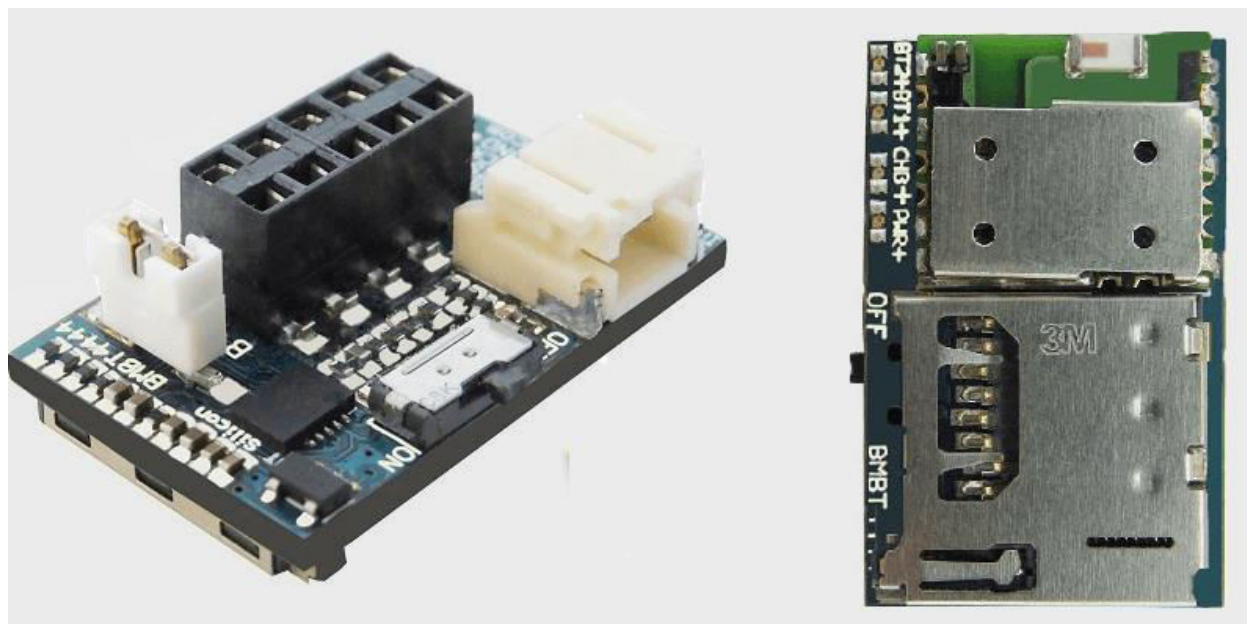


Figure 4: BMBT4444

Feature summary:

- Bluetooth version v3.0
- Micro SD Card Holder*
- Powering option - Li-ion battery, USB
- On-board USB battery charging (with charging current of 200 mA)
- LED indicators for power, battery charging, Bluetooth pairing and communication status
- Power ON-OFF Switch
- JST Connector for Battery
- Jumper to disconnect Bluetooth (*Remove jumper to disconnect Bluetooth.*)
- 10-pins female connector (2.54 mm pitch) with UART, SPI and power IOs
- Dimensions: 26.6 mm x 16.7 mm

*Contact us for firmware support of Micro SD Card.

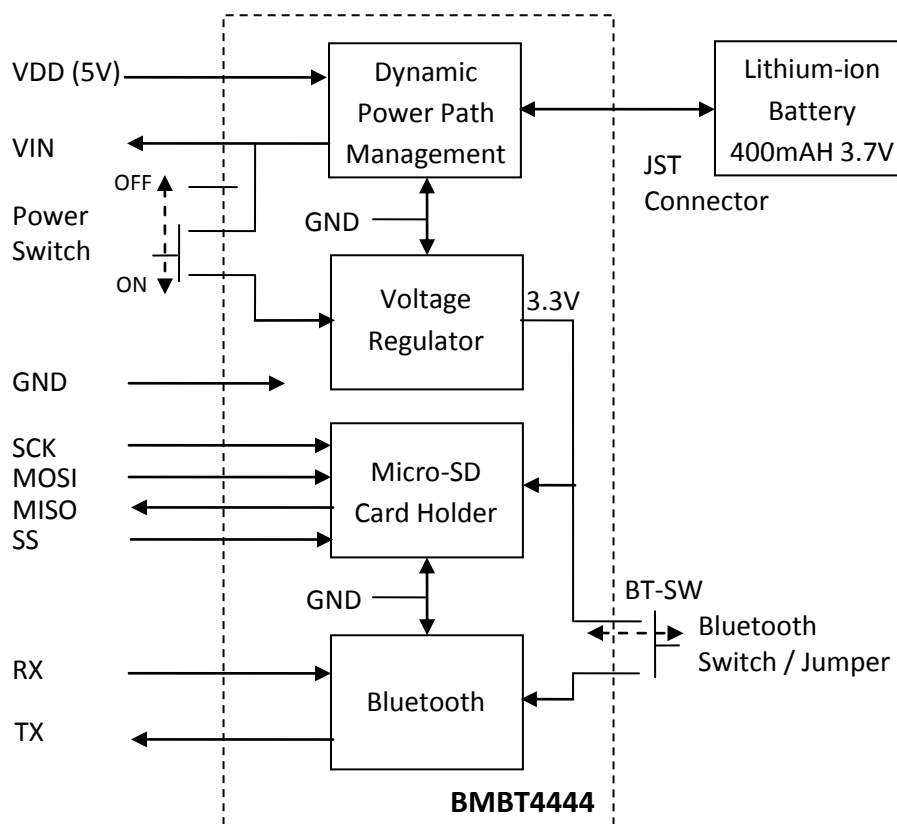


Figure 5: Block diagram of BMBT4444

MIMU48XC + BMBT4444

Mating of MIMU48XC and BMBT4444 results in an integrated device, which is capable of communicating with the application platform without any wire. Data can be transmitted over Bluetooth interface. The integrated device can run on battery and does not require USB power. Depending upon application, Bluetooth module can be selectively disconnected and the data can be store in the onboard micro SD card*. Disconnecting Bluetooth module results in power saving.



Figure 6: MIMU48XC + BMBT4444

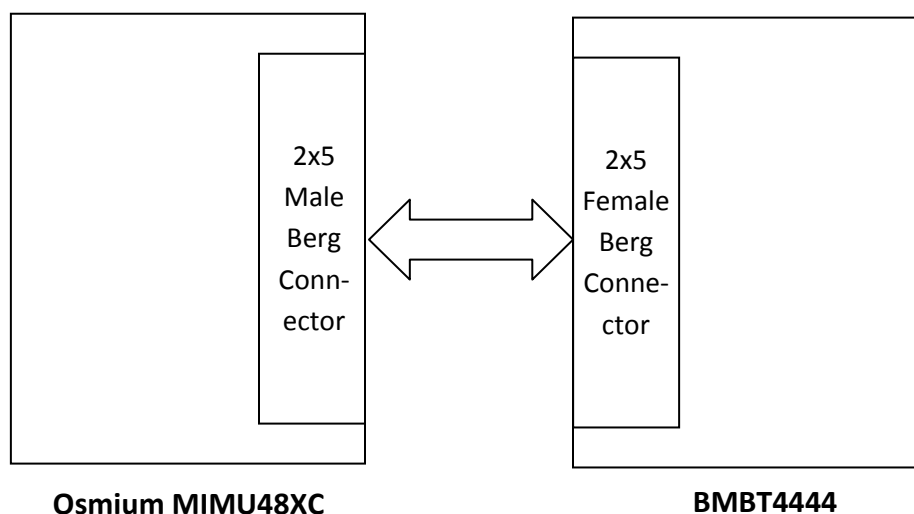


Figure 7: Block Diagram of Integrated Device